

Extended sign-kill for general multi-row $\widehat{\lambda}$: the meta-theorem at arbitrary $\ell(\widehat{\lambda}) \geq 2$

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Abstract

Let $\widehat{\lambda}$ be a partition with $\ell(\widehat{\lambda}) \geq 2$ and every row of length ≥ 2 , and let $\lambda = (\widehat{\lambda}, 1^q) \vdash n = q + |\widehat{\lambda}|$. Set

$$\tau(\widehat{\lambda}) := \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|).$$

We prove that, for $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$, every $T \in \text{supp}(B_{\ell+1}^{(\lambda)} V_\lambda)$ satisfies $p(T) \geq \tau(\widehat{\lambda}) + 1$, and hence

$$B_{\ell+1}^{(\lambda)} V_\lambda \subseteq \bigcap_{j=1}^{\tau(\widehat{\lambda})} E_j^-|_{V_\lambda}.$$

The proof is a strong induction on $|\widehat{\lambda}|$ extending the May-14 two-row meta-theorem. For $\ell(\widehat{\lambda}) = 2$ the refined formula reduces to $\tau = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$ and the May-14 paper is exactly the base layer of the induction. The crucial new ingredient is the *threshold-matching identity* in its multi-row form: every length-preserving contributor $\mu_{k,*}$ has $\tau(\widehat{\mu_{k,*}}) = \tau(\widehat{\lambda})$ exactly, in both the *regular* case ($\widehat{\lambda}_k \geq 3$, $|\widehat{\mu}|$ drops by 1, ℓ unchanged) and the *shrinking* case ($\widehat{\lambda}_k = 2$ at $k = \ell(\widehat{\lambda})$, $|\widehat{\mu}|$ drops by 2, ℓ drops by 1). All non-contributing pieces ($\binom{r}{2}$ corner pairs inside $\widehat{\lambda}$ and all admissible same-row pieces at corners of $\widehat{\lambda}$) have inner shape μ with $\ell(\mu) = \ell(\widehat{\lambda}) + q$ and threshold $\tau(\widehat{\mu}) = \tau(\widehat{\lambda}) + 2$, so the IH provides $B^{(\mu)} \subseteq E_1^-|_{V_\mu}$ and the extra leftmost $R_\ell'^{(\mu)}$ factor kills them.

1 Setup

We use the conventions of the May-13 – May-14 series. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ and quadratic relation $T_j^2 = (q-1)T_j + q$. V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$.

For $1 \leq j \leq n-1$, set $S_j := T_j + 1$. Write

$$E_j^+ := \ker(T_j - q)|_{V_\lambda} = \text{Im}(S_j|_{V_\lambda}), \quad E_j^- := \ker(S_j)|_{V_\lambda}.$$

For $k \geq 1$, set $R'_k := S_{k-1}S_{k-2} \cdots S_1$ (with $R'_1 = 1$). For $\nu \vdash N$ with $\ell = \ell(\nu)$,

$$\Omega^{(\nu)} := R_{\ell+1}'^{(\nu)} R_{\ell+2}'^{(\nu)} \cdots R_N'^{(\nu)}, \quad B_{\ell+1}^{(\nu)} V_\nu := \Omega^{(\nu)} V_\nu.$$

We write $\Omega = \Omega^{(\lambda)}$ and $B = B_{\ell+1}^{(\lambda)} V_\lambda$ when no ambiguity.

Notation for $\widehat{\lambda}$ and λ . Throughout, $\widehat{\lambda}$ denotes a partition with $\ell(\widehat{\lambda}) \geq 2$ rows, every row of length ≥ 2 . We write $\widehat{\lambda}_k$ for the length of row k , with $\widehat{\lambda}_{\ell(\widehat{\lambda})+1} := 0$ by convention. We set

$$\lambda := (\widehat{\lambda}_1, \dots, \widehat{\lambda}_{\ell(\widehat{\lambda})}, \underbrace{1, \dots, 1}_q) \vdash n, \quad n = q + |\widehat{\lambda}|, \quad \ell := \ell(\lambda) = \ell(\widehat{\lambda}) + q.$$

We write ℓ for $\ell(\lambda)$, reserving $\ell(\widehat{\lambda})$ for $\ell(\widehat{\lambda})$.

Identity prefix length. For $T \in \text{SYT}(\mu)$,

$$p(T) := \max\{P \geq 0 : T(i, 1) = i \text{ for all } 1 \leq i \leq P\}.$$

For $W \subseteq V_\mu$, $\text{supp}(W) := \{T \in \text{SYT}(\mu) : \exists w \in W, [v_T]w \neq 0\}$.

Removable corners of λ . The removable corners of λ are

- $c_k = (k, \widehat{\lambda}_k)$ for each $k \in K_{\widehat{\lambda}} := \{k : \widehat{\lambda}_k > \widehat{\lambda}_{k+1}\}$ (the removable corners of $\widehat{\lambda}$); and
- $c_* = (\ell, 1) = (\ell(\widehat{\lambda}) + q, 1)$, the bottom of the column-1 chain.

Note that $\ell(\widehat{\lambda}) \in K_{\widehat{\lambda}}$ always. Set $r := |K_{\widehat{\lambda}}|$; we have $1 \leq r \leq \ell(\widehat{\lambda})$.

2 Threshold formula and the key identity

The threshold τ is defined by

$$\tau(\widehat{\lambda}) := \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|). \quad (1)$$

This is a refinement of the two-row formula $\tau = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$ (recovered when $\ell(\widehat{\lambda}) = 2$, since $|\widehat{\lambda}| = \widehat{\lambda}_1 + \widehat{\lambda}_2$). Equivalently, $\tau = \max(0, q - 1 - (|\widehat{\lambda}| - 2\ell(\widehat{\lambda})))$; the quantity $|\widehat{\lambda}| - 2\ell(\widehat{\lambda}) = \sum_{k=1}^{\ell(\widehat{\lambda})} (\widehat{\lambda}_k - 2)$ counts cells of $\widehat{\lambda}$ outside its first two columns.

The proof rests on the following identity, the multi-row generalization of the May-14 *threshold-matching identity* [[threshold-matching-identity]].

Lemma 1 (Threshold matching, multi-row form). *Let $\widehat{\lambda}$ be as above and $\lambda = (\widehat{\lambda}, 1^q)$. Fix $k \in K_{\widehat{\lambda}}$ and let $\mu_{k,*}$ be the partition obtained from λ by removing the cells $c_k = (k, \widehat{\lambda}_k)$ and $c_* = (\ell(\widehat{\lambda}) + q, 1)$. Write $\widehat{\mu_{k,*}}$ for the $\widehat{\cdot}$ -part of $\mu_{k,*}$ (rows of length ≥ 2) and q' for the number of trailing column-1 cells of $\mu_{k,*}$. Then in the formula $\tau(\nu) = q' + 2\ell(\nu') - 1 - |\nu'|$ (with $\nu' = \widehat{\nu}$):*

- (i) Regular case ($\widehat{\lambda}_k \geq 3$). $\widehat{\mu_{k,*}} = (\widehat{\lambda}_1, \dots, \widehat{\lambda}_k - 1, \dots, \widehat{\lambda}_{\ell(\widehat{\lambda})})$, $\ell(\widehat{\mu_{k,*}}) = \ell(\widehat{\lambda})$, $|\widehat{\mu_{k,*}}| = |\widehat{\lambda}| - 1$, $q' = q - 1$.
- (ii) Shrinking case ($\widehat{\lambda}_k = 2$, forcing $k = \ell(\widehat{\lambda})$). $\widehat{\mu_{k,*}} = (\widehat{\lambda}_1, \dots, \widehat{\lambda}_{\ell(\widehat{\lambda})-1})$, $\ell(\widehat{\mu_{k,*}}) = \ell(\widehat{\lambda}) - 1$, $|\widehat{\mu_{k,*}}| = |\widehat{\lambda}| - 2$, $q' = q$.

In both cases,

$$q' + 2\ell(\widehat{\mu_{k,*}}) - 1 - |\widehat{\mu_{k,*}}| = q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|.$$

That is, if we momentarily denote the un-floored threshold by $\tau_0(\widehat{\lambda}) := q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|$, then $\tau_0(\widehat{\mu_{k,*}}) = \tau_0(\widehat{\lambda})$.

Proof. We use the unfloored quantity throughout. Both cases are direct substitution.

Case (i): $\widehat{\lambda}_k \geq 3$. Row k of $\widehat{\lambda}$, after losing one cell, still has length $\widehat{\lambda}_k - 1 \geq 2$ and remains in $\widehat{\mu}_{k,*}$. The number of rows of $\widehat{\mu}_{k,*}$ is therefore $\ell(\widehat{\lambda})$; its cell count is $|\widehat{\lambda}| - 1$ (one cell removed); and one cell was removed from the tail by deleting c_* , so $q' = q - 1$. Substituting:

$$\tau_0(\widehat{\mu}_{k,*}) = (q - 1) + 2\ell(\widehat{\lambda}) - 1 - (|\widehat{\lambda}| - 1) = q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}| = \tau_0(\widehat{\lambda}).$$

Case (ii): $\widehat{\lambda}_k = 2$. Since $\widehat{\lambda}$ is a partition with all rows ≥ 2 and $\widehat{\lambda}_k = 2$, the only way for c_k to be removable in $\widehat{\lambda}$ is $\widehat{\lambda}_{k+1} \leq \widehat{\lambda}_k - 1 = 1$. By the assumption $\widehat{\lambda}_{k+1} \geq 2$ (for $k + 1 \leq \ell(\widehat{\lambda})$), this forces $k = \ell(\widehat{\lambda})$. Removing c_k leaves row k of length 1, which is no longer in $\widehat{\mu}_{k,*}$ but becomes the topmost cell of the column-1 tail of $\mu_{k,*}$. Counting: $\widehat{\mu}_{k,*}$ has $\ell(\widehat{\lambda}) - 1$ rows; $|\widehat{\mu}_{k,*}| = |\widehat{\lambda}| - 2$ (the corner cell and the demoted cell both leave $\widehat{\lambda}$); and the tail picks up the demoted cell while losing c_* , so $q' = q$. Substituting:

$$\tau_0(\widehat{\mu}_{k,*}) = q + 2(\ell(\widehat{\lambda}) - 1) - 1 - (|\widehat{\lambda}| - 2) = q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}| = \tau_0(\widehat{\lambda}). \quad \square$$

Remark 2. For pieces in which both removed cells lie inside $\widehat{\lambda}$, an analogous computation shows τ_0 *increases* by 2, not preserved. Specifically, for the corner-pair $\mu_{i,j}$ ($i < j$, both in $K_{\widehat{\lambda}}$) and for the same-row piece $\mu_h^{(k)}$ at corner c_k with $\widehat{\lambda}_k \geq 3$ and $\widehat{\lambda}_k \geq \widehat{\lambda}_{k+1} + 2$: $\tau_0(\widehat{\mu}_X) = \tau_0(\widehat{\lambda}) + 2$. The proof is by the same direct computation; we tabulate the cases in Lemma 11 below.

3 Input from prior papers

We collect the inputs that drive the induction.

Proposition 3 (Support rule, May-14 fourth pass). *Let $W \subseteq V_\nu$ be any subspace, $1 \leq j \leq |\nu| - 1$. If every $T \in \text{supp}(W)$ has j and $j + 1$ in the same column of T , then $W \subseteq E_j^-|_{V_\nu}$.*

Remark 4 (Prefix bound implies sign-kill). If every $T \in \text{supp}(W)$ has $p(T) \geq P$, then for every $1 \leq j \leq P - 1$ the entries $j, j + 1$ sit at column-1 rows $j, j + 1$, sharing column 1. By Proposition 3, $W \subseteq E_j^-$ for $1 \leq j \leq P - 1$.

Lemma 5 (Outer-factor preservation, May-14 fifth pass). *Let $W \subseteq V_\nu$ be a subspace such that every $T \in \text{supp}(W)$ satisfies $p(T) \geq P$. Then for every $j \geq P + 1$, every $T'' \in \text{supp}((T_j + 1)W)$ also satisfies $p(T'') \geq P$.*

Theorem 6 (Phase A endpoint, May-19). *For any H_q -module V and any $1 \leq j \leq N - 1$, $(T_j + 1) \cdots (T_1 + 1)V = E_j^+(V)$. In particular, $R'_N V_\nu = E_{N-1}^+|_{V_\nu}$ where $N = |\nu|$.*

Theorem 7 (Two-row meta-theorem, May-14 twelfth pass). *Let $\widehat{\mu} = (a, b)$ with $a \geq b \geq 2$ and $\mu = (a, b, 1^{q'}) \vdash N$ with $q' \geq a + b - 2$. Then every $T' \in \text{supp}(B_{q'+3}^{(\mu)} V_\mu)$ satisfies $p(T') \geq q' + 4 - a - b$. Consequently, $B_{q'+3}^{(\mu)} V_\mu \subseteq E_j^-|_{V_\mu}$ for $1 \leq j \leq q' + 3 - a - b$.*

Theorem 8 (Hook column-1, May-12). *For $\mu = (k_h, 1^{m_h}) \vdash N$ with $k_h \geq 2$ and $m_h \geq \max(k_h, 2)$, $B_{\ell_0+1}^{(\mu)} V_\mu \subseteq E_1^-|_{V_\mu}$ where $\ell_0 = \ell(\mu) = m_h + 1$.*

Theorem 9 (Hook support lemma, May-14 fifth pass). *For $\mu = (k_h, 1^{m_h}) \vdash N$ with $k_h \geq 2$ and $m_h \geq k_h$, every $T \in \text{supp}(B_{\ell_0+1}^{(\mu)} V_\mu)$ satisfies $p(T) \geq m_h - k_h + 2$.*

The two-row meta-theorem is the case $\ell(\widehat{\lambda}) = 2$ of the theorem we are proving, and provides both the *base case* of the induction (at $|\widehat{\lambda}| = 4$, i.e. $\widehat{\lambda} = (2, 2)$) and the principal source of IH inputs whenever an inner shape happens to have $\ell(\widehat{\lambda}) = 2$.

4 Decomposition of $E_{n-1}^+|_{V_\lambda}$

The $+q$ -eigenspace $E_{n-1}^+|_{V_\lambda}$ is indexed by ways of placing $\{n-1, n\}$ in λ at non-column-adjacent cells. As in the two-row paper, we split these into Φ -pieces.

Contributor pieces. For each $k \in K_{\widehat{\lambda}}$, the pair $\{c_k, c_*\}$ gives a piece $\Phi_{k,*}(V_{\mu_{k,*}})$ with $\ell(\mu_{k,*}) = \ell - 1$.

Vanisher pieces (corner pair). For each unordered pair $\{i, j\}$ with $i, j \in K_{\widehat{\lambda}}$, $i < j$, the pair $\{c_i, c_j\}$ gives a piece $\Phi_{i,j}(V_{\mu_{i,j}})$ with $\ell(\mu_{i,j}) = \ell$.

Vanisher pieces (same-row). For each $k \in K_{\widehat{\lambda}}$ with $\widehat{\lambda}_k \geq 3$ and $\widehat{\lambda}_k \geq \widehat{\lambda}_{k+1} + 2$, the same-row pair $\{(k, \widehat{\lambda}_k - 1), (k, \widehat{\lambda}_k)\}$ gives a piece $\Phi_{h^{(k)}}(V_{\mu_h^{(k)}})$ with $\ell(\mu_h^{(k)}) = \ell$.

Lemma 10 (Decomposition of E_{n-1}^+). *With the above notation, define embeddings $\Phi_X : V_{\mu_X} \hookrightarrow V_\lambda$ on the seminormal basis as follows:*

- For corner-pair pieces $X \in \{(k, *), (i, j)\}$:

$$\Phi_X(v_{T'}) := v_{T_X^+} + \tau_X v_{T_X^-},$$

where T_X^+, T_X^- are the two SYTs of λ obtained from T' by placing $\{n-1, n\}$ at $\{c_X^{(1)}, c_X^{(2)}\}$ in the two possible orders, and $\tau_X \in \mathbb{Q}(q)^\times$ is the Hoefsmit 2-block coefficient at the corresponding axial distance.

- For same-row pieces $X = h^{(k)}$: $\Phi_X(v_{T''}) := v_{T'''}$, where T''' is the unique extension of T'' that places $n-1$ at $(k, \widehat{\lambda}_k - 1)$ and n at $c_k = (k, \widehat{\lambda}_k)$.

Then each Φ_X is injective and T_i -equivariant for $1 \leq i \leq n-3$. The Φ_X images have pairwise disjoint seminormal supports and span $E_{n-1}^+|_{V_\lambda}$ in direct sum, matching $R'_n V_\lambda$ by Theorem 6:

$$E_{n-1}^+|_{V_\lambda} = \bigoplus_{k \in K_{\widehat{\lambda}}} \Phi_{k,*}(V_{\mu_{k,*}}) \oplus \bigoplus_{\substack{i, j \in K_{\widehat{\lambda}} \\ i < j}} \Phi_{i,j}(V_{\mu_{i,j}}) \oplus \bigoplus_{\substack{k \in K_{\widehat{\lambda}} \\ \widehat{\lambda}_k \geq 3, \\ \widehat{\lambda}_k \geq \widehat{\lambda}_{k+1} + 2}} \Phi_{h^{(k)}}(V_{\mu_h^{(k)}}).$$

Proof. This is the standard E_{n-1}^+ decomposition. Injectivity and T_i -equivariance for $i \leq n-3$ follow from the fact that s_i moves only entries $i, i+1 \leq n-2$, both inside the inner μ_X -cells, and Hoefsmit coefficients depend only on inner cells; same-row pieces are degenerate 2-blocks (axial distance 1) that reduce to a single basis vector with no Hoefsmit coefficient.

Support disjointness: each piece fixes a distinct unordered cell pair for $\{n-1, n\}$, listed in the decomposition above.

The dimension count $\dim E_{n-1}^+|_{V_\lambda} = \sum_X \dim V_{\mu_X}$ is the standard branching identity for Hecke modules under removable 2-cell configurations (Hoefsmit, Murphy seminormal form). The particular case of $\lambda = (\widehat{\lambda}, 1^q)$ is recovered piece by piece in the May-14 series for $\ell(\widehat{\lambda}) = 2$, and extends verbatim to arbitrary $\ell(\widehat{\lambda})$ since the relevant local geometry of $\{n-1, n\}$ placement depends only on the surrounding cells of λ , which are entirely determined by $\widehat{\lambda}$ in rows $\leq \ell(\widehat{\lambda})$ and by the column-1 chain below. $\square$ $\square$

We now record the threshold computation for vanisher pieces.

Lemma 11 (Threshold inflation on vanishers). *With notation as in Lemma 10, every vanisher piece $\mu \in \{\mu_{i,j} : i < j\} \cup \{\mu_h^{(k)}\}$ satisfies*

$$\tau_0(\widehat{\mu}) = \tau_0(\widehat{\lambda}) + 2.$$

Moreover $\ell(\widehat{\mu}) \geq \ell(\widehat{\lambda}) - 1$ in all cases.

Proof. Direct case analysis. Set $\tau_0(\widehat{\lambda}) = q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|$.

(a) Corner pair $\mu_{i,j}$, $i < j$ both in $K_{\widehat{\lambda}}$.

- If $\widehat{\lambda}_i \geq 3$ and $\widehat{\lambda}_j \geq 3$: both rows survive in $\hat{\cdot}$. $\widehat{\mu} = (\widehat{\lambda}_1, \dots, \widehat{\lambda}_i - 1, \dots, \widehat{\lambda}_j - 1, \dots)$, $\ell(\widehat{\mu}) = \ell(\widehat{\lambda})$, $|\widehat{\mu}| = |\widehat{\lambda}| - 2$, $q' = q$. Then $\tau_0(\widehat{\mu}) = q + 2\ell(\widehat{\lambda}) - 1 - (|\widehat{\lambda}| - 2) = \tau_0(\widehat{\lambda}) + 2$.
- If $\widehat{\lambda}_j = 2$ (forcing $j = \ell(\widehat{\lambda})$, by the same argument as in Lemma 1(ii)): row j drops from $\hat{\cdot}$ on removing c_j . $\widehat{\mu} = (\widehat{\lambda}_1, \dots, \widehat{\lambda}_i - 1, \dots, \widehat{\lambda}_{\ell(\widehat{\lambda})-1})$, $\ell(\widehat{\mu}) = \ell(\widehat{\lambda}) - 1$, $|\widehat{\mu}| = |\widehat{\lambda}| - 3$ (we lose the corner cell at c_i , the corner cell at c_j , and the demoted cell at row j), $q' = q + 1$ (the demoted cell joins the tail). Then $\tau_0(\widehat{\mu}) = (q + 1) + 2(\ell(\widehat{\lambda}) - 1) - 1 - (|\widehat{\lambda}| - 3) = q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}| + 2 = \tau_0(\widehat{\lambda}) + 2$.

(b) Same-row $\mu_h^{(k)}$ at c_k with $\widehat{\lambda}_k \geq 3$, $\widehat{\lambda}_k \geq \widehat{\lambda}_{k+1} + 2$.

- If $\widehat{\lambda}_k \geq 4$: row k survives in $\hat{\cdot}$ with new length $\widehat{\lambda}_k - 2 \geq 2$. $\ell(\widehat{\mu}) = \ell(\widehat{\lambda})$, $|\widehat{\mu}| = |\widehat{\lambda}| - 2$, $q' = q$. Same as the first sub-case of (a): $\tau_0(\widehat{\mu}) = \tau_0(\widehat{\lambda}) + 2$.
- If $\widehat{\lambda}_k = 3$: row k drops from $\hat{\cdot}$ on removing the pair (new length 1). $\ell(\widehat{\mu}) = \ell(\widehat{\lambda}) - 1$, $|\widehat{\mu}| = |\widehat{\lambda}| - 3$ (the two same-row cells and the demoted cell), $q' = q + 1$. Same as the second sub-case of (a): $\tau_0(\widehat{\mu}) = \tau_0(\widehat{\lambda}) + 2$.

In all cases, $\tau_0(\widehat{\mu}) = \tau_0(\widehat{\lambda}) + 2$ and $\ell(\widehat{\mu}) \in \{\ell(\widehat{\lambda}) - 1, \ell(\widehat{\lambda})\} \geq \ell(\widehat{\lambda}) - 1$. □ □

5 The reduction theorem

Theorem 12 (General reduction). *For $\lambda = (\widehat{\lambda}, 1^q)$ with $\ell(\widehat{\lambda}) \geq 2$ and $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$,*

$$B = \left(\prod_{j=\ell}^{n-2} (T_j + 1) \right) \cdot \sum_{k \in K_{\widehat{\lambda}}} \Phi_{k,*} (B_{\ell(\mu_{k,*})+1}^{(\mu_{k,*})} V_{\mu_{k,*}}). \quad (2)$$

The outer product runs in increasing- j order from left ($j = \ell$) to right ($j = n - 2$).

Proof. We apply the May-13/May-20 pull-through to each summand of Lemma 10.

Step 1 (pulling R' -blocks through Φ). Decompose $R'_{n-k} = (T_{n-k-1} + 1)U_k$ with $U_k = (T_{n-k-2} + 1) \cdots (T_1 + 1)$. Since $T_1, \dots, T_{n-k-3}$ commute with T_{n-1} and the factor T_{n-k-2} acts inside the μ_X -block on each Φ_X image (via T_i -equivariance, $i \leq n - 3$), U_k preserves $E_{n-1}^+|_{V_{\widehat{\lambda}}}$ and $U_k \Phi_X(v) = \Phi_X(R'_{n-k-1}^{(\mu_X)} v)$. Iterating for $k = 1, 2, \dots, |\widehat{\lambda}| - 1$:

$$\Omega \Phi_X(V_{\mu_X}) = \left(\prod_{j=\ell}^{n-2} (T_j + 1) \right) \cdot \Phi_X(Q_{\mu_X} V_{\mu_X}), \quad (3)$$

where $Q_{\mu_X} := R'_{\ell}^{(\mu_X)} R'_{\ell+1}^{(\mu_X)} \cdots R'_{n-2}^{(\mu_X)}$ is a product of $|\widehat{\lambda}| - 1$ inner R' -factors at indices $\ell, \ell + 1, \dots, n - 2 = |\mu_X|$.

Step 2 (contributor pieces are $B_{\ell(\mu)+1}^{(\mu)}$). For a contributor $\mu = \mu_{k,*}$, $k \in K_{\hat{\lambda}}$: $\ell(\mu_{k,*}) = \ell - 1$, so $\ell(\mu_{k,*}) + 1 = \ell$. The factor indices of Q_μ run from ℓ to $|\mu| = n - 2$, which is exactly $\ell(\mu) + 1, \dots, |\mu|$. Hence $Q_\mu = \Omega^{(\mu)}$ and

$$I_{k,*} := Q_{\mu_{k,*}} V_{\mu_{k,*}} = B_{\ell(\mu_{k,*})+1}^{(\mu_{k,*})} V_{\mu_{k,*}}.$$

Step 3 (vanisher pieces are killed). For a vanisher $\mu \in \{\mu_{i,j}\} \cup \{\mu_h^{(k)}\}$: $\ell(\mu) = \ell$, so $\ell(\mu) + 1 = \ell + 1$, while the Q_μ factors start at index ℓ . Hence Q_μ has one extra $R_\ell^{(\mu)}$ factor in the leftmost position, beyond $\Omega^{(\mu)}$:

$$Q_\mu = R_\ell^{(\mu)} \Omega^{(\mu)}.$$

The rightmost factor of $R_\ell^{(\mu)}$ is $(T_1 + 1) = S_1$.

Claim: For each vanisher μ , the inductive hypothesis (proved in Section 6 at all strictly smaller $|\hat{\mu}|$) gives

$$B_{\ell(\mu)+1}^{(\mu)} V_\mu \subseteq E_1^- |V_\mu|.$$

Consequently $S_1 \cdot \Omega^{(\mu)} V_\mu = 0$, so $R_\ell^{(\mu)} \Omega^{(\mu)} V_\mu = 0$, i.e., $Q_\mu V_\mu = 0$.

Justification of the claim. By Lemma 11, $\tau_0(\hat{\mu}) = \tau_0(\hat{\lambda}) + 2$. The multi-row stability hypothesis $q \geq |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2$ gives $\tau_0(\hat{\lambda}) \geq 1$, hence $\tau_0(\hat{\mu}) \geq 3$.

Stability check for IH applicability. The IH at μ requires $q'' \geq |\hat{\mu}| - 2\ell(\hat{\mu}) + 2$, where q'' is the tail count of μ . By Lemma 11 we have the four sub-cases:

- (1) (Corner pair, $\hat{\lambda}_i, \hat{\lambda}_j \geq 3$) or (same-row, $\hat{\lambda}_k \geq 4$): $|\hat{\mu}| = |\hat{\lambda}| - 2$, $\ell(\hat{\mu}) = \ell(\hat{\lambda})$, $q'' = q$. Stability $q \geq |\hat{\lambda}| - 2\ell(\hat{\lambda})$ follows from $q \geq |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2$.
- (2) (Corner pair, $\hat{\lambda}_j = 2$, $j = \ell(\hat{\lambda})$) or (same-row, $\hat{\lambda}_k = 3$): $|\hat{\mu}| = |\hat{\lambda}| - 3$, $\ell(\hat{\mu}) = \ell(\hat{\lambda}) - 1$, $q'' = q + 1$. Stability $q + 1 \geq |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 1$ follows from $q \geq |\hat{\lambda}| - 2\ell(\hat{\lambda})$.

In all sub-cases the IH precondition holds. Since $|\hat{\mu}| \in \{|\hat{\lambda}| - 2, |\hat{\lambda}| - 3\}$ is strictly smaller than $|\hat{\lambda}|$, the inductive hypothesis is well-defined. We split on $\ell(\hat{\mu})$.

- If $\ell(\hat{\mu}) \geq 2$: the inductive hypothesis applies and gives $B_{\ell(\mu)+1}^{(\mu)} V_\mu \subseteq E_j^-$ for $1 \leq j \leq \tau(\hat{\mu}) = \max(0, \tau_0(\hat{\mu}))$. Since $\tau_0(\hat{\mu}) \geq 3 \geq 1$, this includes $j = 1$.
- If $\ell(\hat{\mu}) = 1$: then $\hat{\mu}$ is a single-row partition, and μ is a hook $\mu = (\hat{\mu}_1, 1^{q'})$. We apply Theorem 8 directly to obtain $B^{(\mu)} \subseteq E_1^- |V_\mu|$. The hypothesis $q' \geq \max(\hat{\mu}_1, 2)$ is checked in each sub-case as follows. Length-1 vanisher inner shapes arise only when $\ell(\hat{\lambda}) = 2$: the corner-pair $\mu_{i,j}$ has $\ell(\hat{\mu}) \geq \ell(\hat{\lambda}) - 1$, so $\ell(\hat{\mu}) = 1$ forces $\ell(\hat{\lambda}) = 2$; similarly the same-row $\mu_h^{(k)}$ has $\ell(\hat{\mu}) \geq \ell(\hat{\lambda}) - 1$. In the $\ell(\hat{\lambda}) = 2$ case, the regime check matches verbatim the May-14 two-row meta-paper (Theorem 3.1, Step 3 therein), which we cite as a black box. For $\ell(\hat{\lambda}) \geq 3$, this sub-case is vacuous.

In either sub-case, the claim follows.

Step 4 (assembly). By Lemma 10, $R'_n V_\lambda = E_{n-1}^+ |V_\lambda| = \bigoplus_X \Phi_X(V_{\mu_X})$. Apply $R'_{\ell+1} \cdots R'_{n-1}$ to both sides and decompose via Step 1. Contributors give $\Phi_{k,*}(I_{k,*})$ (Step 2); vanishers give 0 (Step 3). This yields (2). $\square$ $\square$

6 Proof of the main theorem

Theorem 13 (Main). *Let $\widehat{\lambda}$ be a partition with $\ell(\widehat{\lambda}) \geq 2$ rows, every row of length ≥ 2 , and let $\lambda = (\widehat{\lambda}, 1^q) \vdash n$ with $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$. Set $\tau_0(\widehat{\lambda}) := q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|$ and $\tau(\widehat{\lambda}) := \max(0, \tau_0(\widehat{\lambda}))$. Then every $T \in \text{supp}(B)$ satisfies*

$$p(T) \geq \tau_0(\widehat{\lambda}) + 1 = q + 2\ell(\widehat{\lambda}) - |\widehat{\lambda}|. \quad (4)$$

Consequently,

$$B \subseteq \bigcap_{j=1}^{\tau(\widehat{\lambda})} E_j^-|_{V_\lambda}.$$

Proof. Strong induction on $|\widehat{\lambda}|$.

Base case ($|\widehat{\lambda}| = 4$). The only partition with $\ell(\widehat{\lambda}) \geq 2$, every row ≥ 2 , and $|\widehat{\lambda}| = 4$ is $\widehat{\lambda} = (2, 2)$. By the two-row meta-theorem (Theorem 7), every $T \in \text{supp}(B)$ satisfies $p(T) \geq q + 4 - 4 = q$ when $q \geq 2$. This matches $\tau_0(\widehat{\lambda}) + 1 = q + 4 - 1 - 4 + 1 = q$.

Inductive step ($|\widehat{\lambda}| \geq 5$). Fix $\widehat{\lambda}$ with $\ell(\widehat{\lambda}) \geq 2$ and $|\widehat{\lambda}| \geq 5$, and assume the theorem holds for every $\widehat{\mu}$ with $\ell(\widehat{\mu}) \geq 2$, every row ≥ 2 , $|\widehat{\mu}| < |\widehat{\lambda}|$, and $q'' \geq |\widehat{\mu}| - 2\ell(\widehat{\mu}) + 2$. Assume further that for hook inner shapes (length-1 $\widehat{\mu}$), the hook column-1 theorem (Theorem 8 of the May-14 meta-paper) provides $B^{(\mu)} \subseteq E_1^-$.

We must show every $T \in \text{supp}(B)$ has $p(T) \geq \tau_0(\widehat{\lambda}) + 1$.

By Theorem 12,

$$B = \left(\prod_{j=\ell}^{n-2} (T_j + 1) \right) \cdot \sum_{k \in K_{\widehat{\lambda}}} \Phi_{k,*} (B_{\ell(\mu_{k,*})+1}^{(\mu_{k,*})} V_{\mu_{k,*}}).$$

Step 1 (prefix bound on contributor inner pieces). We treat each $k \in K_{\widehat{\lambda}}$ separately and apply Lemma 1.

Regular case ($\widehat{\lambda}_k \geq 3$). Inner shape $\widehat{\mu}_{k,*}$ has $\ell(\widehat{\mu}_{k,*}) = \ell(\widehat{\lambda}) \geq 2$, $|\widehat{\mu}_{k,*}| = |\widehat{\lambda}| - 1 < |\widehat{\lambda}|$, $q' = q - 1$.

- Stability check: $q' \geq |\widehat{\mu}_{k,*}| - 2\ell(\widehat{\mu}_{k,*}) + 2 = (|\widehat{\lambda}| - 1) - 2\ell(\widehat{\lambda}) + 2 = |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 1$. Our hypothesis gives $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$, hence $q' = q - 1 \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 1$. ✓
- Inductive hypothesis applies (since $|\widehat{\mu}| < |\widehat{\lambda}|$). Every $T' \in \text{supp}(B_{\ell(\mu_{k,*})+1}^{(\mu_{k,*})} V_{\mu_{k,*}})$ satisfies $p(T') \geq \tau_0(\widehat{\mu}_{k,*}) + 1 = \tau_0(\widehat{\lambda}) + 1$ (Lemma 1(i)).

Shrinking case ($\widehat{\lambda}_k = 2$, so $k = \ell(\widehat{\lambda})$). Inner shape $\widehat{\mu}_{k,*}$ has $\ell(\widehat{\mu}_{k,*}) = \ell(\widehat{\lambda}) - 1$, $|\widehat{\mu}_{k,*}| = |\widehat{\lambda}| - 2 < |\widehat{\lambda}|$, $q' = q$.

- Stability check: $q' \geq |\widehat{\mu}_{k,*}| - 2\ell(\widehat{\mu}_{k,*}) + 2 = (|\widehat{\lambda}| - 2) - 2(\ell(\widehat{\lambda}) - 1) + 2 = |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$. Our hypothesis gives $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2 = q'$, satisfying the bound with equality at $q = |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$. ✓
- If $\ell(\widehat{\lambda}) - 1 \geq 2$ (i.e., $\ell(\widehat{\lambda}) \geq 3$): the inductive hypothesis applies. Every $T' \in \text{supp}(B_{\ell(\mu_{k,*})+1}^{(\mu_{k,*})} V_{\mu_{k,*}})$ satisfies $p(T') \geq \tau_0(\widehat{\mu}_{k,*}) + 1 = \tau_0(\widehat{\lambda}) + 1$ (Lemma 1(ii)).
- If $\ell(\widehat{\lambda}) - 1 = 1$ (i.e., $\ell(\widehat{\lambda}) = 2$, so $\widehat{\lambda} = (\widehat{\lambda}_1, 2)$ and $\mu_{k,*}$ is a hook): we invoke the May-14 two-row meta paper's handling of the $b = 2$ contributor: $\mu_{k,*} = (\widehat{\lambda}_1, 1^q)$ is a hook with $k_h = \widehat{\lambda}_1$, $m_h = q$. Stability $q \geq \widehat{\lambda}_1$ from $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2 = \widehat{\lambda}_1$. The hook support lemma (Theorem 9 of the

May-14 meta-paper, citing May-14 fifth pass) gives every $T' \in \text{supp}(B_{\ell(\mu_{k,*})+1}^{(\mu_{k,*})} V_{\mu_{k,*}})$ satisfies $p(T') \geq m_h - k_h + 2 = q - \widehat{\lambda}_1 + 2$. This equals $\tau_0(\widehat{\lambda}) + 1$ since $\tau_0(\widehat{\lambda}) = q + 4 - 1 - (\widehat{\lambda}_1 + 2) = q - \widehat{\lambda}_1 + 1$. In all sub-cases, $p(T') \geq \tau_0(\widehat{\lambda}) + 1$.

Step 2 (prefix bound on $\Phi_{k,*}$ -extensions). The Phi-embedding $\Phi_{k,*}$ produces SYTs of λ from SYTs of $\mu_{k,*}$ by adding $\{n-1, n\}$ at the cell pair $\{c_k, c_*\}$.

- Cell $c_k = (k, \widehat{\lambda}_k)$ lies in row k of $\widehat{\lambda}$ at column $\widehat{\lambda}_k \geq 2$, so it is *not* in column 1; it does not disturb the column-1 prefix of the inner SYT.
- Cell $c_* = (\ell(\widehat{\lambda}) + q, 1)$ in λ sits at row $\ell(\widehat{\lambda}) + q = \ell$, strictly below row $\ell(\mu_{k,*}) = \ell - 1$ where all of $\mu_{k,*}$'s column-1 cells live. So c_* is appended strictly below the prefix-relevant region.

Therefore for any Phi-extension $T \in \text{supp}(\Phi_{k,*}(W))$, $T(j, 1) = T'(j, 1)$ for $1 \leq j \leq \ell - 1$, and in particular $p(T) \geq p(T')$. Combining with Step 1,

$$p(T) \geq p(T') \geq \tau_0(\widehat{\lambda}) + 1.$$

Step 3 (outer factors preserve the prefix). The outer factors are $(T_j + 1)$ for $\ell \leq j \leq n - 2$. The hypothesis of Lemma 5 requires $j \geq p(T') + 1 = \tau_0(\widehat{\lambda}) + 2 = q + 2\ell(\widehat{\lambda}) - |\widehat{\lambda}| + 1$. The smallest outer index is $j = \ell = q + \ell(\widehat{\lambda})$, so we need

$$q + \ell(\widehat{\lambda}) \geq q + 2\ell(\widehat{\lambda}) - |\widehat{\lambda}| + 1, \quad \text{i.e.,} \quad |\widehat{\lambda}| \geq \ell(\widehat{\lambda}) + 1.$$

This holds since every row of $\widehat{\lambda}$ has length ≥ 2 , so $|\widehat{\lambda}| \geq 2\ell(\widehat{\lambda}) \geq \ell(\widehat{\lambda}) + 1$ for $\ell(\widehat{\lambda}) \geq 1$. ✓ Applying Lemma 5 along the $|\widehat{\lambda}| - \ell(\widehat{\lambda}) - 1$ outer factors in sequence, the prefix bound $\geq \tau_0(\widehat{\lambda}) + 1$ is preserved.

Conclusion. By Steps 1–3, every $T \in \text{supp}(B)$ satisfies $p(T) \geq \tau_0(\widehat{\lambda}) + 1$.

Under the regime hypothesis $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$ we have $\tau_0(\widehat{\lambda}) \geq 1$, so $\tau(\widehat{\lambda}) = \tau_0(\widehat{\lambda})$ and by Remark 4 the prefix bound $p(T) \geq \tau_0(\widehat{\lambda}) + 1$ yields $B \subseteq E_j^-$ for $1 \leq j \leq p(T) - 1 \geq \tau_0(\widehat{\lambda}) = \tau(\widehat{\lambda})$. This completes the induction. $\square$

Remark 14 (Recursive structure). The induction descends $|\widehat{\lambda}|$ by 1 at each contributor recursion (regular case) and by 2 at each contributor recursion (shrinking case). Vanisher recursions descend $|\widehat{\lambda}|$ by 2 or 3. The induction is well-founded with base $|\widehat{\lambda}| = 4$.

Remark 15 (Threshold matching). The combinatorial heart of the proof is the threshold-matching identity (Lemma 1): every length-preserving contributor preserves τ_0 exactly, in both the regular and shrinking cases. Without this identity the prefix bound on the contributor would not lift cleanly to the prefix bound on B . In the May-14 two-row meta-paper, the analogous identity was $\tau(\widehat{\mu}_{i,3}) = \tau(\widehat{\lambda})$; here it generalizes to multi-row via the refined formula $\tau_0 = q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|$.

7 Subsumed cases and new ground covered

Theorem 13 subsumes the May-14 two-row meta-theorem (as the base layer $\ell(\widehat{\lambda}) = 2$) and extends it to all higher row counts.

Table 1: Multi-row cases. Rows above the line are covered by the May-14 two-row meta-theorem; rows below are new with the present theorem.

$\widehat{\lambda}$	$ \widehat{\lambda} $	$\ell(\widehat{\lambda})$	$q \geq q_0$	status
$(2, 2), \dots, (a, b)$	$\leq \infty$	2	$a + b - 2$	May-14 meta (two-row)
$(2, 2, 2)$	6	3	2	new ($\ell(\widehat{\lambda}) = 3$ base)
$(3, 2, 2)$	7	3	3	new
$(3, 3, 2), (4, 2, 2)$	8	3	4	new
$(3, 3, 3), (4, 3, 2), (5, 2, 2)$	9	3	5	new
$(2, 2, 2, 2)$	8	4	2	new ($\ell(\widehat{\lambda}) = 4$ base)
$(3, 2, 2, 2)$	9	4	3	new
$\vdots$				
$(\widehat{\lambda}_1, \dots, \widehat{\lambda}_\ell(\widehat{\lambda})), \ell(\widehat{\lambda}) \geq 2$	$ \widehat{\lambda} $	$\ell(\widehat{\lambda})$	$ \widehat{\lambda} - 2\ell(\widehat{\lambda}) + 2$	new (general multi-row)

8 Computational verification

The empirical support is the 24-shape data assembled in `~/projects/scratch/2026-05-15-multi-row-tau/`: twenty shapes with $\widehat{\lambda}$ having a length-2 bottom row (`results.md`), and four additional shapes at $\ell(\widehat{\lambda}) \in \{3, 4\}$ that distinguish the multi-row refinement (`run_hard_shapes.out`). The script `multi_row_tau.py` computes $\dim B$ and the actual threshold τ_{act} for each shape by: (i) Hoefsmit seminormal construction of V_λ at $Q = 11 \bmod P = 100003$; (ii) sparse application of $\Omega = R'_{\ell+1}^{(\lambda)} \cdots R'_n^{(\lambda)}$ to random vectors; (iii) rank stability check across 60–300 probes; (iv) $\tau_{\text{act}} = \max\{j : (T_j + 1)\Omega v = 0 \text{ for 3 random } v\}$ and sharpness at $\tau_{\text{act}} + 1$.

$\widehat{\lambda}$	$\ell(\widehat{\lambda})$	$ \widehat{\lambda} $	range of q tested	predicted τ_0	observed τ_{act}
$(3, 2, 2)$	3	7	$2 \leq q \leq 5$	$q - 2$	$\max(0, q - 2) \checkmark$
$(4, 2, 2), (3, 3, 2)$	3	8	$2 \leq q \leq 5$	$q - 3$	$\max(0, q - 3) \checkmark$
$(4, 3, 2), (3, 2, 2, 2)$	3, 4	9	$2 \leq q \leq 5$	$q - 4, q - 2$	matches $\checkmark$
$(4, 3, 2, 2)$	4	11	$2 \leq q \leq 3$	$q - 4$	0 $\checkmark$
$(4, 3^2, 1^5), (3^4, 1^4)$	3, 4	≥ 10	specific q	≥ 1	$\geq 1 \checkmark$

The 24 datapoints have been verified to match $\tau_{\text{act}} = \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|)$ and to be sharp at $\tau_{\text{act}} + 1$.

9 Discussion

9.1 What this proves

The containment direction of the multi-row extended sign-kill conjecture for *all* $\widehat{\lambda}$ with $\ell(\widehat{\lambda}) \geq 2$ and every row ≥ 2 , uniformly, at the refined threshold $\tau_0(\widehat{\lambda}) = q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|$ and the multi-row stability bound $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$.

9.2 What this does not prove

Sharpness. $B \not\subseteq E_{\tau+1}^-$ is not addressed structurally. Empirically, all 24 multi-row datapoints exhibit sharpness at $\tau + 1$, consistent with the two-row evidence.

The non-stable regime. For $q < |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$ the behavior is not described by this theorem; in two-row work, the non-stable regime exhibits non-semisimplicity phenomena (May-14 non-stable/JM paper) and the dim formula and threshold both fail naively.

The intermediate cell-vs-row distinction. The `deep-row-results.md` note observes that the formulas $\tau_0 = q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|$ and $\tau_0 = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2 - r$ (with $r = \#\{k \geq 3 : \widehat{\lambda}_k \geq 3\}$) cannot be distinguished by data on shapes with $\widehat{\lambda}_3 \geq 4$ at currently feasible q . However, the proof above establishes the *containment* direction at the cell-count threshold $\tau_0 = q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|$, which is a strictly stronger statement than $\tau_0 = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2 - r$ when they differ. So this theorem also resolves the cell-vs-row ambiguity in the containment direction.

9.3 The trace-vanishing form

The companion wake-2 finding [[trace-vanishing-conjecture]] observed that $\text{tr}_{V^\lambda}(\Omega) = 0$ iff $\tau(\widehat{\lambda}) \geq 1$ on six test shapes. Theorem 13 gives the *containment* side of this in one direction: if $B \subseteq E_1^-$ then $T_1\Omega|_B = -\Omega|_B$, hence $\text{tr}(T_1\Omega) = -\text{tr}(\Omega)$ (restricted to B); the trace-vanishing conjecture is the integrated form. The other direction (trace vanishing implies $\tau \geq 1$) remains a separate question.

10 Status

Established.

- Theorem 13: $B \subseteq E_j^-$ for $1 \leq j \leq \tau(\widehat{\lambda}) = \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|)$, for all $\widehat{\lambda}$ with $\ell(\widehat{\lambda}) \geq 2$ and every row ≥ 2 , and all $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$.
- Subsumes the May-14 two-row meta-theorem as the base layer $\ell(\widehat{\lambda}) = 2$.
- Extends extended sign-kill to all multi-row shapes empirically tested.

Open.

- Structural sharpness at $j = \tau(\widehat{\lambda}) + 1$ for multi-row $\widehat{\lambda}$.
- Trace-vanishing in the reverse direction.
- The non-stable regime $q < |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$.
- Larger- q computational verification on shapes with $\widehat{\lambda}_3 \geq 4$, where the formula's exact form (cell vs row) becomes testable.